# KRUSHI-CONNECT

FROM FARMERS' HANDS TO YOURS

FINAL VIVA PROJECT PRESENTATION

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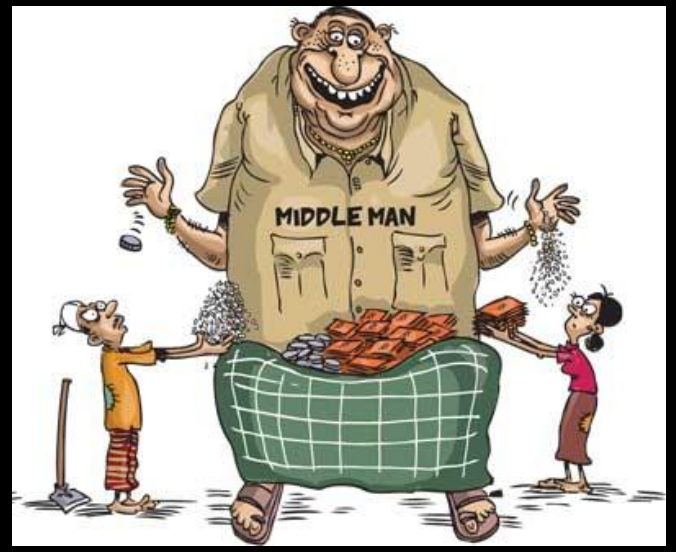
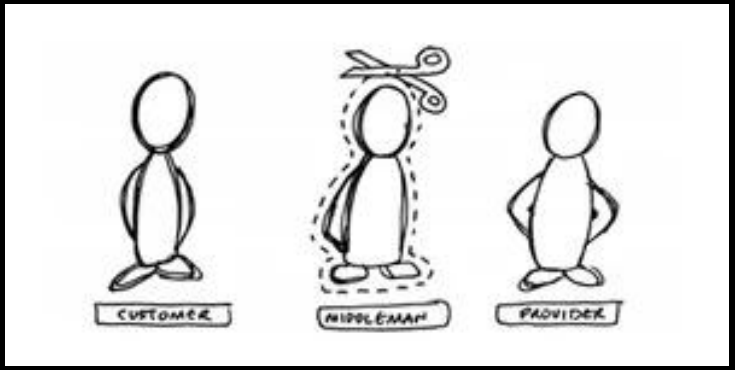
# Problem Statement

- 75% of Sri Lankan farmers are small-scale and rely on middlemen. (FAO)
- Middlemen take 30%-50% of the profit, reducing farmer earnings. (CBSL)
- Post-harvest losses due to inefficient supply chains range from 30%-40%. (MoA)
- Limited digital adoption - only ~20% of farmers use online selling due to lack of awareness and accessibility. (ICTA)
- Consumers face higher prices and limited access to fresh farm produce. (SLCAA)



# Motivation




- Eliminate middlemen → Ensure fair pricing for farmers.
- Direct-to-Consumer (D2C) model → More profits for farmers, fresher produce for consumers.
- User-friendly digital platform → Supports the needs of all user types involved.
- Reduce waste → Optimize farm-to-market logistics.

# Key Objectives

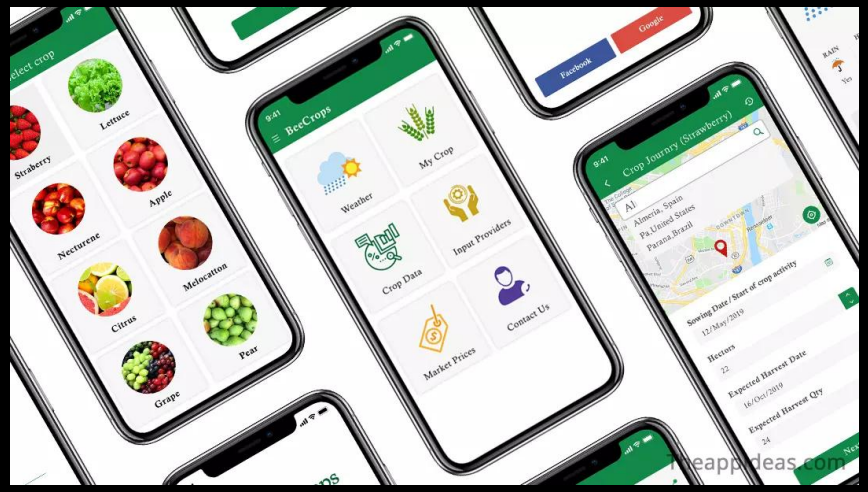
- **Facilitate Direct Farmer-to-Consumer Sales:** Eliminate middlemen to increase farmer earnings.
- **Enable Location-Based Services:** Utilize Google Maps externally for product location tracking and customer access on pickup orders.
- **Provide Secure Authentication & Role-Based Access:** Implement Firebase authentication for farmers, customers, and admins (with Email link authorization and fingerprint integration for admin related access).
- **Order Management & Transactions:** Enhanced order processing, and payments integration.
- **Telephony Services:** Allow customers to directly call farmers for inquiries.
- **Receipt Downloading:** Provide an option to download order receipts for record-keeping.
- **Implement Product Filtering & Search:** Allow users to filter by category, price, and location for better product discovery.

# Technologies & Architecture

## Frontend (Android - Java & XML) :

- Technologies Used:
  - XML: UI Layouts.
  - Android Design Palette: Manual UI structuring.
  - Java: Manages UI logic (Activities, Fragments, Adapters, ViewModel).

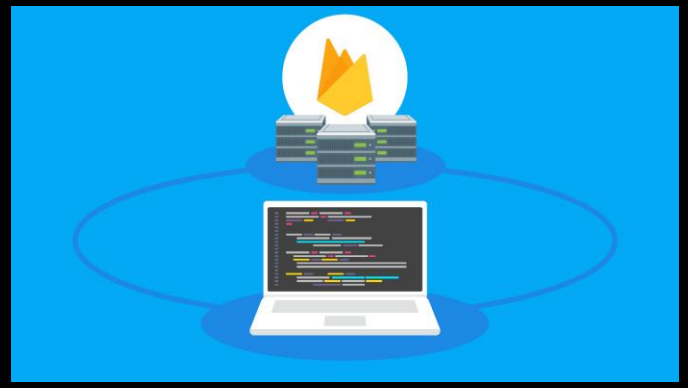
- Architecture Used:
  - MVVM (Model-View-ViewModel)
    - Model (M): Handles data-related operations (Firestore models, authentication models).
    - View (V):
      - XML Layouts: UI design for activities & fragments.
      - Android Design Palette: Drag-and-drop UI structuring.
    - ViewModel (VM): Manages UI logic and API interactions.



# Technologies & Architecture

## Backend (Firebase Services & APIs)

- Technologies Used:
  - Java: Handles backend logic and Firestore interactions.
  - Firestore (NoSQL Database): Stores application related data.
  - Firebase Authentication: Role-based login/signup management.
  - Firebase Storage: Stores product images.Firestore Security.
- Network & API Handling:
  - Java (Firestore Queries & Data Processing): Manages Firestore CRUD operations.
  - Java (FirebaseAuth & Firebase Storage SDK): Handles authentication and image uploads.
- APIs & External Integrations:
  - Google Maps & Places APIs: Fetches and displays locations  Telephony: Allows direct calls.
  - PayHere: Processes payments securely.



# Key Features : Farmers

- Farmer Sign-Up & Authentication:
  - Only farmers (role) can register and login using farmer signup & sign-in pages.
- Product Management:
  - Farmers can add products (only farmers) and update stocks of their products.
  - Uploads support three images per product via Firebase storage.
  - Admin approval required for farmer products to be listed on the app.
- Order Management:
  - Farmers can view only their own ordered products and gets to update the status on pickup orders of customers.
- Location Services:
  - Product location fetched dynamically from the farmer's stored address.
- Direct Customer Interaction:
  - Customers can call farmers directly via the Telephony API from order details.

# Key Features : Customers

- Product Browsing & Ordering:
  - Customers can browse, search and filter on conditions through admin-approved products.
  - Location-Based Product Filtering with customers being able to view nearby farmers' products on Google Maps externally.
  - Customers can add products to the cart and place orders with secure payments through PayHere.
  - Customers are able to download a receipt related to their order.
- After Ordering:
  - Customers can check order status directly in the app as they are updated by farmers and admin.
  - UI updates dynamically.
- Account Management: Customer can update their information they entered when registering (password also - if needed).

# Key Features : Admin

- Admin Dashboard:
  - Centralized product and order management.
- Product Approval System:
  - Admins review products marked as "Pending Approval“ as products can be approved (made visible to customers) or rejected.
- Order Monitoring: Admins have full control over order status updates.
- Security Features for Admins:
  - Password less Email Authentication for Login
  - When accessing critical application process data fingerprint authorization is needed (biometric access control).

# Conclusion & Takeaways

- What was achieved?
  - A fully functional agricultural marketplace connecting farmers & customers.
- Benefits of Krushi-Connect:
  - For Farmers → Easy product listing, direct customer interaction, increased sales visibility.
  - For Customers → Seamless access to fresh, farm-direct products, real-time farmer contact.
  - For Admins → Efficient platform management with secure control over products, and orders.
- Scalability:
  - The platform is designed for growth, with the upcoming enhancements such as AI-based pricing suggestions for farmers, Real-time chat support, Real-time delivery support, Multi-Language Support and more payment methods integrated.

# THANK-YOU

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